PHASE THEORY — PAPER 43

Lorentz Invariance as Emergent Phase Symmetry

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Abstract

Phase Theory proposes that phase is the sole fundamental constituent of physical reality. In this paper — Paper 43 of the Phase Theory series — we derive Lorentz invariance as an emergent symmetry of admissible phase propagation, not as a fundamental spacetime postulate. Without assuming spacetime, metrics, light cones, or relativistic kinematics as primitives, we show that Lorentz symmetry arises whenever admissible phase reconfiguration admits a universal maximum propagation rate and isotropic admissibility geometry. Relativistic effects — time dilation, length contraction, relativistic mass-energy behavior, and invariant intervals — emerge as structural consequences of phase ordering constraints. We formalize this derivation through 12 propositions and 6 definitions (beyond those carried from prior papers), compare Phase Theory's treatment of Lorentz symmetry against standard special relativity, loop quantum gravity, causal set theory, and string-theoretic approaches, identify the precise regime conditions under which Lorentz symmetry admits phase-native deviations, and enumerate five classes of experimental predictions. This paper demonstrates that Einstein's postulates are effective summaries of deeper phase-native structure, and that spacetime does not enforce Lorentz symmetry — phase demands it.

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1. Why Lorentz Invariance Cannot Be Assumed

The principle of Lorentz invariance occupies a foundational position in modern theoretical physics. In Einstein's special theory of relativity [1], it enters as a postulate: the laws of physics, and in particular the speed of light in vacuum, are identical in all inertial reference frames. In quantum field theory, Lorentz invariance is imposed as a fundamental symmetry of the action, and the entire representational apparatus of particle physics — spinors, four-vectors, the Lorentz group SO(1,3) and its covering group SL(2,C) — is constructed to be manifestly Lorentz-covariant [6]. The Standard Model Lagrangian is, to extraordinary experimental precision, Lorentz-invariant. Lorentz invariance is not merely a successful empirical hypothesis; it is, in the standard view, a structural feature of the universe at the most fundamental level accessible to current physics.

Yet this posture conceals a deep foundational tension. Assuming Lorentz invariance as a primitive postulate is epistemically unsatisfying precisely because it forecloses the question of why the universe should possess this symmetry. A theory that accepts Lorentz invariance without derivation is, in the philosopher's sense, explanatorily incomplete: it describes the symmetry but does not account for it. This incompleteness is tolerable as a practical matter when the goal is computing scattering amplitudes or predicting particle masses, but it becomes critically problematic in the context of foundational theories that aim to describe the origin of spacetime itself. If spacetime is emergent — if it is not a primitive manifold but a structure arising from some deeper substrate — then Lorentz invariance cannot be assumed at the level of that substrate. It must be derived from the properties of whatever is more fundamental.

The problem is sharpest in quantum gravity. Loop quantum gravity, causal set theory, spin foam models, and various discrete spacetime approaches all confront the question of how Lorentz symmetry can be preserved or recovered when the underlying structure is discrete or non-manifold. The answers vary, and none are entirely satisfactory [19, 24, 25]. String theory avoids the problem by assuming Lorentz invariance in ten or twenty-six dimensions from the outset [20], but this merely relocates the postulate rather than eliminating it. Emergent gravity approaches [15, 16] take spacetime to arise from thermodynamic or entropic considerations, yet still assume special relativity as a boundary condition. The logical gap is consistent: every major approach to quantum gravity or emergent spacetime either assumes Lorentz invariance or must import it from some prior level of description without derivation.

Attempts to derive Lorentz invariance from deeper principles have a long history. Ignatowski [4] showed in 1910 that Lorentz transformations can be derived from assumptions of linearity, isotropy, homogeneity, the relativity principle, and group structure — without independently postulating a universal speed. Bacry and Lévy-Leblond [5] in 1968 classified all kinematic groups compatible with standard symmetry assumptions and showed that Lorentz invariance is essentially unique among them. Pelissetto and Testa [22] in 2015 demonstrated a clean derivation of Lorentz transformations from isotropy, the relativity principle, and causality. These are mathematically elegant results. However, they share a critical limitation: all of them operate within the conceptual framework of spacetime. They derive Lorentz transformations as transformations of spacetime coordinates and thus implicitly presuppose that spacetime exists as an arena within which these transformations act. They do not address the question of why that arena should have Lorentzian rather than Galilean or Euclidean geometry.

Phase Theory's approach is ontologically distinct. Phase Theory [27] proposes that the sole primitive constituent of physical reality is phase: a relational, substrate-level structure from which all geometric, kinematic, and dynamical concepts emerge. Spacetime itself is emergent in Phase Theory [28] — it is not a pre-existing manifold but a description of ordered phase relations that becomes applicable in the coherent propagation regime. The logical order is therefore reversed relative to all previous approaches: in Phase Theory, phase comes first, spacetime is derived from phase, and Lorentz symmetry is derived from the properties of the phase substrate. There is no level at which Lorentz invariance is assumed; it is a theorem, not an axiom.

This paper is dedicated to making that derivation complete, rigorous, and self-contained. We will show that three conditions on the phase substrate — bounded admissible propagation rate, isotropy of the admissibility geometry, and absence of preferred orientation — are individually derivable from Phase Theory's three foundational axioms (Paper 1 [27]), and that these three conditions jointly and necessarily imply Lorentz symmetry in the emergent spacetime description. The Lorentz group SO(1,3) is not installed by hand; it falls out of the structure of admissible phase reconfiguration as inevitably as Euclidean geometry falls out of the axioms of Hilbert's geometry. The present paper, Paper 43 of the Phase Theory series, provides the most complete treatment of this derivation to date, incorporating and systematizing results first indicated in Papers 1, 10, and 38 [27, 28, 29].

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2. Review of Phase Theory Foundations Relevant to This Paper

Phase Theory rests on three foundational axioms, stated in Paper 1 [27] and carried forward throughout the series. We restate them here for completeness, in the precise form required for the derivations in this paper. The **Phase Primacy Axiom** asserts that phase — conceived as a relational property of the substrate elements of reality — is the unique ontological primitive. No other entity, including particles, fields, spacetime points, or wavefunction amplitudes, is primitive; all are effective descriptions of phase structure valid in appropriate regimes. The **Global Consistency Axiom** asserts that the phase substrate must at all times and in all regions satisfy a global consistency condition: no admissible reconfiguration of any local region of the substrate can produce a global inconsistency. This axiom functions as the phase-theoretic analogue of a constraint equation and drives the emergence of conservation laws, causal structure, and symmetry. The **Admissibility Axiom** asserts that not all phase configurations are equally accessible: there exists a functional — the admissibility function — that assigns to each proposed phase reconfiguration a value in [0,1] measuring the degree to which that reconfiguration is consistent with the global constraint.

The phase substrate is denoted S. Formally, S is a set whose elements φ are called phase loci. Phase loci are not points in space; they are not events in spacetime; they are not quantum states. They are the pre-geometric constituents from which all such structures emerge. The admissibility function is denoted $A[\varphi]: S \times S \to [0, 1]$, where $A[\varphi_1, \varphi_2]$ measures the admissibility of the transition from locus φ_1 to locus φ_2 in a single update cycle. An admissible phase reconfiguration is any sequence of transitions $\{\varphi_0 \to \varphi_1 \to \varphi_2 \to \ldots\}$ such that every consecutive pair has $A[\varphi_i, \varphi_{i+1}] > 0$. The set of all admissible phase reconfigurations from a given initial locus constitutes the reachability structure of that locus, and it is from this structure that causal and temporal order emerge.

Phase Theory distinguishes between two modes of phase behavior. The *kinematic mode* refers to phase configurations that propagate smoothly through the substrate, maintaining coherence over extended regions. The *topological mode* refers to phase configurations that involve non-trivial winding numbers or defects — configurations that cannot be continuously deformed to the kinematic mode. Phase defects, studied extensively in Paper 1 [27], are topological-mode objects and serve as the phase-theoretic analogues of matter. Their admissibility depth, studied in Paper 38 [29], is the phase-theoretic analogue of inertia and rest mass.

Spatial ordering, temporal ordering, and metric geometry all emerge from phase relations, as derived in Paper 10 [28]. We recall the key results. Spatial ordering emerges from phase adjacency: two phase loci are spatially adjacent if and only if their mutual admissibility $A[\varphi_1, \varphi_2]$ exceeds a threshold ε_s. The emergent spatial distance is the shortest admissible path length in units of adjacency steps. Temporal ordering emerges from phase succession: locus φ_2 is temporally later than locus φ_1 if and only if φ_2 is reachable from φ_1 by an admissible reconfiguration but not vice versa. This defines the temporal partial order $\prec$ on S, which — in the coherent propagation regime — generates a well-defined temporal dimension. The emergent metric geometry in the coherent regime is a Lorentzian manifold, but this fact is derived, not assumed; the derivation of Lorentz signature is the primary task of the present paper.

The admissibility depth function $D[\varphi]$, introduced in Paper 38 [29], quantifies the resistance of a phase configuration to reconfiguration. Formally, $D[\varphi]$ is the minimum number of admissible update cycles required to transform φ into any configuration in its topological equivalence class that differs from it by a prescribed amount. For kinematic-mode configurations, D is small and nearly constant. For topological defects (matter analogues), D is large and plays the role of inertial mass. The behavior of D under velocity — i.e., under phase translation at rate v relative to the ambient substrate ordering — is the subject of Section 10, where we derive relativistic mass-energy behavior from first principles.

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3. The Admissibility Geometry — Formal Structure

We now introduce the formal geometric structure of the admissibility function, which will be the central object of the derivations that follow. The admissibility geometry encodes, at each phase locus, the directions and rates at which phase reconfiguration can proceed. Understanding its structure is prerequisite to deriving Lorentz invariance.

Definition 43.1 (Admissibility Geometry). For a given phase locus $\varphi \in S$, the *admissibility geometry* at φ is the collection of all admissible phase reconfiguration vectors at φ, together with the rate structure imposed on them by the admissibility function A. Formally, let $T_\varphi S$ denote the set of infinitesimal phase displacements at φ (in the limit of continuous phase approximation). The admissibility geometry assigns to each direction $d \in T_\varphi S$ a maximum admissible propagation rate $r(d) = \sup\{ v : A[\varphi, \varphi + v\cdot d\cdot\delta t] > 0 \text{ as } \delta t \to 0 \}$. The *admissibility cone* at φ is defined as $C(\varphi) = \{ (d, v) \in T_\varphi S \times R_{\geq 0} : v \leq r(d) \}$, the set of all direction-rate pairs that are admissible at φ.

Definition 43.2 (Local Isotropy). The admissibility cone $C(\varphi)$ is said to be *locally isotropic* at locus φ if the function $r: T_\varphi S \to R_{\geq 0}$ is invariant under the emergent spatial rotation group SO(3) at φ. That is, $r(d) = r(Rd)$ for all $R \in SO(3)$ and all $d \in T_\varphi S$. Equivalently, the cross-section of $C(\varphi)$ at any fixed rate $v \leq r_{max}$ is a sphere in the emergent spatial directions, exhibiting full SO(3) symmetry.

Definition 43.3 (Global Consistency Enforcement). The *global*

> *consistency operator* $\hat{G}$ is the operator acting on the phase substrate S that enforces the Global Consistency Axiom. Formally, $\hat{G}$ maps any proposed phase configuration Φ to the unique admissible configuration closest to Φ in the admissibility metric. When Φ is already admissible, $\hat{G}\Phi = \Phi$. When Φ violates global consistency, $\hat{G}\Phi$ is the nearest admissible configuration, which in general differs from Φ by a non-local phase rearrangement.

The global consistency operator $\hat{G}$ plays a crucial role in constraining local phase dynamics. The key theorem linking global consistency to local admissibility is the following.

> **Theorem (Global-Local).** Global consistency enforcement by $\hat{G}$ implies that every local admissibility cone $C(\varphi)$ must satisfy the boundedness condition: there exists a finite positive constant r_{max} such that $r(d) \leq r_{max}$ for all $d \in T_\varphi S$ and all $\varphi \in S$.

Proof. Suppose, for contradiction, that r(d) is unbounded for some direction d at some locus φ_0. Then admissible phase influence can propagate from φ_0 in direction d at arbitrarily large rate. But unbounded propagation rate implies that the influence of a local reconfiguration at φ_0 can reach arbitrarily distant loci in arbitrarily short phase-time. The Global Consistency Axiom requires that any reconfiguration be consistent with the entire substrate S; if phase influence propagates without bound, then any local reconfiguration immediately imposes constraints on the entire substrate simultaneously. This is possible only if the substrate has zero admissibility depth everywhere, which contradicts the fact that S supports topological phase defects with finite admissibility depth (Paper 38 [29]). The contradiction establishes that $r(d)$ must be bounded above by a finite r_{max} for all d and all φ. □

The relationship between the admissibility geometry and the emergent metric is established by the following correspondence. In the coherent propagation regime (far from admissibility saturation), the admissibility cone $C(\varphi)$ is smooth, convex, and locally constant across neighboring loci. The propagation metric on S, defined in Appendix A, is then well-approximated by a smooth Lorentzian metric on the emergent spacetime manifold M derived from S. The metric signature is determined by the asymmetry between the temporal ordering (which is a strict partial order on S, hence one-dimensional) and the spatial adjacency structure (which is isotropic and three-dimensional in the physical regime). We show formally in Appendix A that these conditions yield signature $(-, +, +, +)$.

The proof that the admissibility geometry is Lorentzian when the isotropy and boundedness conditions hold is the core mathematical result of this paper. In brief: if $C(\varphi)$ is isotropic ($SO(3)$-symmetric) in the spatial directions and bounded by r_{max} in all directions, and if the temporal direction is distinguished by the irreversibility of the phase ordering $\prec$, then the set of all (direction, rate) pairs (d, v) with $v = r_{max}$ forms a codimension-1 surface in $T_\varphi S$ that, by isotropy, is a sphere in the spatial directions and a single point in the temporal direction. The set of all admissible propagation four-vectors with $v = r_{max}$ therefore traces out the null cone of a Lorentzian metric with v_{max} playing the role of c. Phase influence propagating at $v < r_{max}$ corresponds to timelike vectors; at $v = r_{max}$ to null vectors; and no admissible reconfiguration can correspond to spacelike propagation ($v > r_{max}$) by the boundedness theorem above.

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4. Definition of Maximum Admissible Phase Rate

We now formalize the concept of the maximum admissible phase propagation rate, which will serve as the phase-theoretic correlate of the speed of light. This quantity is not assumed; it is derived from the properties of the admissibility function and the global consistency constraint.

> **Definition 43.4** (Maximum Admissible Phase Rate). The *maximum admissible phase rate* v_{max} is the finite supremum of all admissible phase reconfiguration propagation speeds across emergent spatial ordering. Formally,
>
> (43.1) $v_{max} = \sup \{ v \in \mathbb{R}_{\geq 0} : \exists\, d \in T_\varphi S, A[\varphi, \varphi + v \cdot d \cdot \delta t] > 0$ for arbitrarily small $\delta t > 0 \}$.
>
> By the Global-Local Theorem of Section 3, v_{max} is finite. By the non-triviality of the admissibility function (which must admit at least one admissible transition for the substrate to support any dynamics), v_{max} is strictly positive.

The derivation of v_{max} from the ratio of phase update density to admissibility constraint density proceeds as follows. Let ρ_{update} denote the maximum number of admissible phase updates per unit phase-time at a given locus, and let $\rho_{constraint}$ denote the admissibility constraint density — the number of independent consistency conditions per unit emergent spatial volume imposed by the Global Consistency Axiom. Then the rate at which admissible phase influence can advance through the substrate in a given spatial direction is limited by the requirement that each step of the advance must satisfy all local admissibility constraints before the next step can proceed. This serialization constraint yields the dimensional relation

$$\text{(43.1a)}\quad v_{max} \sim \rho_{update} / \rho_{constraint},$$

from which it follows that v_{max} is finite whenever $\rho_{constraint}$ is nonzero and finite — which it must be in any substrate that supports a non-trivial global consistency condition. In the coherent propagation regime, v_{max} coincides with the empirical speed of light c to within the precision of current experiments (see Section 15). The distinction between v_{max} and c becomes observable only near the admissibility saturation scale (Section 16).

> **Remark 43.1.** The quantity v_{max} is not a postulate of Phase Theory. It is a consequence of the Global Consistency Axiom and the Admissibility Axiom. Its finiteness follows from Theorem 3 (Global-Local); its uniqueness follows from the isotropy of the admissibility geometry (Section 5). v_{max} does not depend on the choice of phase locus φ in the coherent regime — this frame-independence is derived, not assumed, and corresponds exactly to Einstein's postulate that the speed of light is the same in all inertial frames.

Comparison with Planck-scale cutoffs in other frameworks is instructive. In doubly special relativity (DSR) [21], both the speed of light and the Planck energy are postulated as frame-independent invariants, leading to modified dispersion relations. In loop quantum gravity, the Planck length acts as a minimum spatial resolution below which the spin network description applies [17]. In Phase Theory, neither v_{max} nor the admissibility saturation scale ℓ_{sat} is postulated; both emerge from the structure of A[φ]. The admissibility saturation scale is the spatial scale at which $\rho_{constraint}$ approaches its maximum possible value ρ_{max}, beyond which no further admissible update density is possible. We derive this scale and its observational consequences in Section 16.

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5. Isotropy of the Admissibility Geometry — Proof

The second key ingredient in the derivation of Lorentz invariance is the isotropy of the admissibility geometry. We now derive this isotropy from the Phase Theory axioms, establishing that there is no preferred spatial direction at the phase-substrate level.

Definition 43.5 (Phase Orientation). A *preferred orientation* on the phase substrate S is a nontrivial vector field V on S — a mapping $\varphi \to V(\varphi) \in T_\varphi S$, with $V(\varphi) \neq 0$ for a set of loci of positive measure — such that V is preserved under all admissible reconfigurations: if $\varphi \to \varphi'$ is admissible, then the parallel transport of $V(\varphi)$ along the reconfiguration path equals $V(\varphi')$. Equivalently, a preferred orientation is a non-trivial global section of the tangent bundle of S that is invariant under the admissibility flow.

Lemma 43.1. The admissibility geometry at any locus $\varphi \in S$ is locally isotropic if and only if the phase substrate S admits no preferred orientation at φ.

Proof. (⇒) Suppose the admissibility geometry is isotropic at φ. Then $r(d) = r(Rd)$ for all $R \in SO(3)$. If a preferred orientation $V(\varphi)$ existed, there would be a distinguished direction $d_V = V(\varphi)/|V(\varphi)|$ such that $r(d_V) \neq r(d)$ for generic d, violating isotropy. Contradiction.

(⇐) Suppose no preferred orientation exists. Then for any two spatial directions d, d' at φ there exists an admissibility-preserving rotation R with $Rd = d'$. Since R preserves the admissibility function (no preferred direction exists to break rotational symmetry), we have $r(d) = r(Rd) = r(d')$. Since d, d' are arbitrary, r is constant on the unit sphere in $T_\varphi S$, which is exactly the definition of local isotropy. □

Proposition 43.1. Phase Theory's Global Consistency Axiom forbids preferred orientations on the phase substrate S.

Proof. Suppose a preferred orientation V exists on S. V is a non-trivial vector field preserved under all admissible reconfigurations. Consider the following: since V is non-trivial, there exists a region $R_V \subset S$ in which $V(\varphi)$ points in a definite direction, say d_0, in the emergent coordinate description. Now apply the Global Consistency Axiom: any admissible reconfiguration of S must be globally consistent. Consider the admissible reconfiguration σ that rotates all phase adjacency relations in region R_V by an angle π around an axis perpendicular to d_0. By the rotational symmetry of the admissibility function restricted to spatial substructure (which holds because the admissibility function depends only on relational properties of phase loci, not on any absolute orientation — there is no background against which absolute orientation could be defined in Phase Theory), σ is admissible. But σ maps d_0 to $-d_0$, while V requires d_0 to be preserved. The resulting configuration is inconsistent with the existence of V, violating the invariance of V under admissible reconfigurations. This contradicts the definition of a preferred orientation. Therefore no preferred orientation can exist on S. □

Corollary 43.1. The admissibility geometry is isotropic at every locus $\varphi \in S$ (in the coherent propagation regime), recovering the full spatial rotation group SO(3) as an emergent symmetry of the phase substrate.

Proof. By Proposition 43.1, S admits no preferred orientations. By Lemma 43.1, this implies isotropy of the admissibility geometry at every φ. The emergent rotation group SO(3) is then the symmetry group of the isotropic admissibility cone cross-section at any locus, which is a 2-sphere. The symmetry group of a 2-sphere is SO(3). □

It should be noted that isotropy is a statement about the coherent propagation regime. At the boundary of admissibility saturation — for instance, at the phase genesis event or in extreme-density phase configurations — the admissibility geometry may develop anisotropic features. In the phase genesis era, before global consistency has been established across the full substrate, local anisotropies are possible. These would leave imprints in the earliest cosmological observables, a point we return to in Section 27. In the present section, the proof of Corollary 43.1 is understood to apply within the coherent regime that describes all of observed physics below the admissibility saturation scale.

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6. Emergence of Causal Structure from Bounded Phase Propagation

Definition 43.6 (Phase Causal Cone). For a given phase locus φ ∈ S, the *phase causal cone* J&sup+;(φ) is defined as the set of all phase loci reachable from φ by at least one admissible reconfiguration sequence: J&sup+;(φ) = { ψ ∈ S : ∃ admissible path γ from φ to ψ }. The *future phase* *causal cone* J&sup+;(φ) and *past phase causal cone* J−(φ) are defined analogously, with the temporal ordering relation ≺ distinguishing past from future.

Proposition 43.2. When admissible phase influence propagates at a rate bounded by v_max and the admissibility geometry is isotropic (Corollary 43.1), a well-defined causal cone structure emerges from the phase substrate, with the property that J&sup+;(φ) in emergent coordinates is the forward light cone of a Lorentzian metric with characteristic speed v_max.

Proof. Let φ be a given phase locus, and let t = 0 be the emergent time coordinate assigned to φ by the phase ordering relation ≺. At emergent time t > 0, the set of phase loci reachable from φ by admissible propagation is bounded by the maximum propagation distance v_max · t in any spatial direction. By Corollary 43.1, the admissibility geometry is isotropic, so this bound is the same in every spatial direction. The boundary of J&sup+;(φ) at time t is therefore the sphere of radius v_max · t in emergent space. This is precisely the boundary of the forward light cone {(t, x) : |x| = v_max t, t > 0} of the Lorentzian metric $ds^2 = -v_{max}^2\, dt^2 + dx^2 + dy^2 + dz^2$. The interior of J&sup+;(φ) corresponds to timelike-separated regions (reachable at v < v_max), and no phase influence can reach the exterior (which would require v > v_max, violating Definition 43.4). Therefore the causal cone structure of phase propagation coincides in emergent coordinates with the forward light cone of a Lorentzian metric with c ≡ v_max. □

> **Remark 43.2.** Light cones are not spacetime primitives in Phase Theory. They are level sets of the admissibility propagation function — specifically, the boundary of the phase causal cone J&sup+;(φ). The identification of the boundary ∂J&sup+;(φ) with the null cone of a Lorentzian metric is a theorem (Proposition 43.2), not an assumption. This means that the concept of "null interval" is derived rather than fundamental: it marks the boundary between phase-influence-reachable and phase-influence-unreachable regions.

The emergence of chronological and causal order from phase ordering proceeds as follows. The phase ordering relation $\prec$ on S — defined by reachability under admissible reconfiguration — is a strict partial order: irreflexive ($\varphi \nprec \varphi$), asymmetric ($\varphi \prec \psi$ implies $\psi \nprec \varphi$), and transitive ($\varphi \prec \psi$ and $\psi \prec \xi$ implies $\varphi \prec \xi$). These properties follow directly from the irreversibility built into the admissibility function: if $A[\varphi_1, \varphi_2] > 0$, then in general $A[\varphi_2, \varphi_1] = 0$ in the coherent regime, reflecting the temporal asymmetry of phase dynamics. The emergent chronological order of spacetime events is then the restriction of $\prec$ to pairs of events separated by a timelike phase path, and the causal order is the restriction to pairs separated by any admissible path (timelike or null).

Causal set theory [20, 15] takes as its fundamental postulate that spacetime is a locally finite partial order. Phase Theory's phase ordering $\prec$ resembles this structure but differs in a crucial respect: in causal set theory, the partial order is postulated on a set of spacetime events; in Phase Theory, the partial order is derived from the admissibility structure of a set of pre-geometric phase loci. The dimensionality of the emergent spacetime, which is a deep problem in causal set theory (requiring independent assumptions about sprinkling density), emerges in Phase Theory from the admissibility adjacency structure: three independent spatial directions are possible because the admissibility geometry supports SO(3)-isotropic propagation, and three is the minimum dimension in which SO(3) acts effectively. This point is elaborated in Section 21.

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7. Derivation of Lorentz Transformations from Phase Structure

We now present the central derivation of this paper. The goal is to show that the group of transformations preserving admissible phase propagation is precisely the Lorentz group SO(1,3), derived entirely from the phase-theoretic conditions established in the preceding sections.

> **Theorem 43.1** (Main Theorem). If admissible phase propagation is bounded by a finite rate v_max (Definition 43.4) and isotropic (Corollary 43.1), then the group of bijections T: S → S that preserve the phase causal cone structure — i.e., that map every phase causal cone to a phase causal cone — is isomorphic to the Lorentz group SO(1,3) in the emergent coordinate description, with v_max playing the role of the invariant speed c.

Proof. We proceed in four steps, working in the emergent coordinate description established by Phase Theory's spatial and temporal ordering (Paper 10 [28]).

Step (i): Linearity. The admissibility function A is uniform across the substrate in the coherent propagation regime — this is a consequence of the Global Consistency Axiom, which enforces that no locus has privileged admissibility properties relative to any other. Therefore the propagation metric defined by A is translationally invariant (homogeneous) in emergent coordinates. A bijection preserving the causal cone structure of a homogeneous Lorentzian propagation metric must be an affine map in the emergent coordinates (this follows from the Alexandrov–Zeeman theorem [23], generalized to the phase-substrate setting). Affine maps compose with a linear part and a translation; translations trivially preserve the causal cone structure, so we may focus on the linear part. Let T be a linear map on the emergent coordinate space R&sup4; (with coordinates (t, x, y, z) identifying phase ordering and spatial adjacency).

Step (ii): Hyperbolic Geometry. T must preserve the maximum phase propagation rate v_max. A phase locus ψ is on the boundary of J&sup+;(φ) if and only if its emergent coordinates (t, x, y, z) satisfy $|x|^2 + |y|^2 + |z|^2 = v\_max^2\ t^2$. This is the null cone condition $Q(x) = -v\_max^2\ t^2 + x^2 + y^2 + z^2 = 0$. Preservation of J&sup+;(φ) by T requires preservation of the null cone $Q(x) = 0$. Since T is linear, it must preserve Q up to an overall scaling: $Q(Tx) = \lambda\ Q(x)$ for some $\lambda \neq 0$. But T must also preserve the temporal orientation (future-directed admissible paths map to future-directed admissible paths, by the admissibility axiom), which fixes the sign of λ. Furthermore, the normalization of the phase ordering implies that unit phase-time intervals map to unit phase-time intervals (the definition of the temporal coordinate is tied to the phase update cycle, not to an external clock), which fixes $|\lambda| = 1$ and therefore $\lambda = 1$. Thus T preserves Q exactly: $Q(Tx) = Q(x)$ for all x ∈ R&sup4;.

Step (iii): Spatial Isotropy. By Corollary 43.1, the admissibility geometry is SO(3)-isotropic in the spatial directions. Therefore T must commute with all spatial rotations: $T\ R\_s = R\_s\ T$ for all $R\_s \in SO(3)$ acting on the (x, y, z) subspace. The group of linear maps on R&sup4; that preserve Q and commute with spatial rotations is the group of Lorentz boosts plus spatial rotations. Together with Step (ii), T is constrained to lie in the orthogonal group O(1,3) of the quadratic form Q.

Step (iv): Group Composition. The set of all admissibility-preserving transformations forms a group under composition (composition of two admissibility-preserving maps is admissibility-preserving; the identity is trivially admissibility-preserving; invertibility holds by the bijectivity assumption). By Steps (i)–(iii), this group is a subgroup of O(1,3). Restricting to orientation-preserving (det T = +1) and future-directed (temporal orientation preserved) elements gives the proper orthochronous Lorentz group SO&sup+(1,3) ≅ SO(1,3). □

The explicit transformation formulas are derived from the constraint Q(Tx) = Q(x) with T a boost in the x-direction at emergent velocity v. Writing T as a 2×2 block acting on (t, x), the constraint gives

$$(43.2)\ T = [\gamma \quad -\gamma v/v\_max^2 ; -\gamma v \quad \gamma]$$

where γ is determined by the requirement T&sup{-1} = T(v → −v) and Q(Tx) = Q(x):

$$(43.3)\ \gamma = (1 - v^2/v\_max^2)\&sup\{-1/2\}.$$

The Lorentz factor γ (43.3) has a precise phase-theoretic interpretation: it is the *phase compression ratio*, the factor by which the admissibility budget of a phase defect moving at velocity v is redistributed between internal phase updates (contributing to proper time) and translational phase propagation (contributing to spatial displacement). The explicit transformation formulas for all four coordinates are then

$$(43.4)\ t' = \gamma(t - vx/v\_max^2),$$

$$(43.5)\ x' = \gamma(x - vt),$$

$$(43.6)\ y' = y,$$

$$(43.7)\ z' = z.$$

The velocity addition formula follows from composing two boosts $T(v_1)$ and $T(v_2)$:

$$(43.8)\ v_{1+2} = (v_1 + v_2) / (1 + v_1 v_2 / v_{max}^2),$$

which is bounded above by v_{max} for all $v_1, v_2 < v_{max}$ — a phase-theoretic consequence of the admissibility bound, not a kinematic coincidence.

Corollary 43.2. Einstein's two postulates of special relativity — (i) the principle of relativity (physical laws identical in all inertial frames) and (ii) the invariance of the speed of light — are necessary consequences of Phase Theory's three foundational axioms, not independent assumptions.

Proof. Postulate (ii) follows immediately from Theorem 43.1: v_{max} is frame-independent by Proposition 43.6 (Section 12), and $v_{max} = c$ in the coherent propagation regime. Postulate (i) follows from Proposition 43.6 and Corollary 43.3 (Section 12), which establish that no preferred inertial frame exists and that admissibility laws are identical at every locus. □

The derivation requires no prior notion of inertial frames. Inertial frames in Phase Theory are emergent: they are descriptions of phase configurations in which the local admissibility geometry is well-approximated by the global average geometry. The equivalence of all inertial frames is then a consequence of the homogeneity of the admissibility function in the coherent regime, not an independent postulate.

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8. Time Dilation as Phase Ordering Redistribution

Phase Theory's account of time, developed in Paper 10 [28], identifies temporal succession with the ordering of admissible phase updates at a given phase locus. Each phase update cycle at locus φ constitutes one unit of proper time at φ. The total number of admissible phase updates that a phase defect (matter analogue) undergoes along a given trajectory through the substrate constitutes its elapsed proper time. This identification is not merely metaphorical; it is the precise phase-theoretic definition of proper time, and it immediately motivates the derivation of time dilation as a consequence of admissibility budget redistribution.

> **Proposition 43.3.** When a phase defect moves with velocity v relative to the ambient phase ordering structure, the rate of internal phase updates at the defect is reduced by the factor γ&sup{-1} = $\sqrt{(1 - v^2/v\_max^2)}$ relative to a stationary defect.

Proof. Let B denote the total admissibility budget per phase update cycle available to a phase defect in the coherent propagation regime. B is a fixed finite constant determined by the admissibility function. For a stationary defect (φ fixed in emergent spatial coordinates), all of B is available for internal phase updates (reconfiguration of the defect's internal phase structure), contributing to the progression of proper time at the defect. For a defect translating at velocity v in the x-direction, a fraction of B must be expended on the translational phase transport: each unit of emergent spatial displacement $\Delta x = v\Delta t$ requires a phase reconfiguration cost proportional to v/v_max per update cycle (since v_max is the rate at which one full admissibility-budget unit propagates the maximum spatial distance). By the Lorentzian geometry of the admissibility cone (Theorem 43.1), the translational and internal phase update contributions to B are not additive but related by the Pythagorean relation of the Lorentzian metric: if $\beta = v/v\_max$, then the fraction of B available for internal updates is $\sqrt{(1 - \beta^2)} = \gamma$&sup{-1}. Therefore the proper time rate at a moving defect is γ&sup{-1} times that at a stationary defect. □

This derivation yields the standard time dilation formula:

$$\text{(43.9)}\quad \Delta\tau = \Delta t \sqrt{(1 - v^2/v\_max^2)} = \Delta t / \gamma,$$

where $\Delta\tau$ is the proper time elapsed at the moving defect and Δt is the coordinate time elapsed in the ambient phase ordering frame.

> **Remark 43.3.** This derivation does not require a metric structure, a reference to clocks, or any concept of inertial frames. It follows from the admissibility budget conservation alone. The result is not that clocks slow down: it is that phase update cycles redistribute. What physics calls "time dilation" is in Phase Theory the necessary consequence of the admissibility budget being shared between internal and translational phase processes, exactly as a fixed power supply must divide between two loads.

Phase Theory recovers all standard experimental verifications of time dilation. GPS satellite clock corrections require accounting for both special-relativistic time dilation (velocity-dependent, from Proposition 43.3) and general-relativistic gravitational time dilation (from Section 26's treatment of Phase Theory in the curved-geometry regime). The Hafele–Keating experiment [15] involved atomic clocks flown on aircraft and compared to ground clocks; Phase Theory predicts exactly the observed asymmetric aging, via equation (43.9) for the SR contribution and the phase-theoretic gravitational correction for the GR contribution. The Rossi–Hall experiment on cosmic ray muons confirmed that muons created in the upper atmosphere reach ground-level detectors at rates consistent with time dilation factors of ~10, exactly as Phase Theory predicts from Proposition 43.3 with $\gamma \sim 10$ at the relevant energies.

The twin paradox (treated in detail in Section 25) is resolved in Phase Theory without appeal to acceleration or frame switching. The key insight is that the admissibility budget consumed by a phase defect along its trajectory through the substrate is a path-dependent quantity, analogous to the length of a curve in Riemannian geometry. Two defects following different admissible paths through S from the same starting locus to the same ending locus will in general have consumed different total budgets, and thus completed different numbers of internal update cycles — corresponding to different elapsed proper times. This is an objective, coordinate-independent fact about the phase substrate, not a perspective-dependent statement.

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9. Length Contraction as Phase Adjacency Compression

Spatial extension in Phase Theory is an emergent property of phase adjacency relations, as established in Paper 10 [28]. The spatial distance between two phase loci is the number of adjacency steps (admissible spatial transitions) in the shortest admissible path connecting them. The spatial extent of a phase defect is the number of adjacency steps spanning its internal phase structure in a given emergent spatial direction. Under rapid phase transport — that is, when the defect moves at velocity v relative to the ambient phase ordering — this adjacency structure is modified in a systematic way.

> **Proposition 43.4.** Under rapid phase transport at velocity v in direction

> $\hat{n}$, the number of admissible spatial distinctions along $\hat{n}$ within a phase defect is reduced by the factor γ&sup{-1} = $\sqrt{(1 - v^2/v_{max}^2)}$ relative to the stationary count, while the spatial distinctions in directions perpendicular to $\hat{n}$ are unaffected.

Proof. Consider a phase defect D spanning N_0 adjacency steps in direction $\hat{n}$ when stationary. When D propagates at velocity v in direction $\hat{n}$, each adjacency step in that direction must be re-established in each update cycle. The re-establishment cost per adjacency step in the direction of motion is not zero: the defect's leading edge must propagate at v in $\hat{n}$, while its trailing edge also propagates at v. For the defect to maintain N adjacency steps in $\hat{n}$, each step must survive the update cycle. However, the admissibility geometry (Theorem 43.1) constrains the combined longitudinal and temporal phase resolution: the longitudinal spatial resolution and the temporal resolution are not independent when $v \neq 0$; they are related by the Lorentz geometry of the admissibility cone. Specifically, the longitudinal spatial resolution per update cycle is reduced by factor γ&sup{-1} relative to the stationary case, because the update cycle simultaneously advances the defect by v in $\hat{n}$, consuming a fraction of the adjacency budget in that direction. The transverse directions (perpendicular to $\hat{n}$) are unaffected, because no net transverse transport occurs and the transverse admissibility budget is fully available for spatial distinction (by isotropy, Corollary 43.1). □

This yields the length contraction formula:

$$L' = L / \gamma = L \sqrt{(1 - v^2/v_{max}^2)}, \quad (43.10)$$

where L is the rest length of the defect (spatial extent when stationary) and L' is its moving length in the direction of motion. The transverse dimensions y and z are unchanged, consistent with the isotropy constraint.

The distinction between phase compression (ontological) and measurement artifact (epistemic) is crucial for the correct interpretation of (43.10). Phase Theory holds that length contraction is real: the number of admissible phase adjacency steps constituting the defect in the direction of motion is genuinely reduced. This is not a perspective-dependent effect; it is a structural consequence of the admissibility geometry. However, this does not imply that the internal phase structure of the defect is in any sense "damaged" or "compressed" in a physically meaningful absolute sense — the defect's proper spatial extent (measured in its own rest frame, by the number of adjacency steps in that frame) is unchanged. The contraction is entirely a consequence of the admissibility geometry's behavior under relative motion, not a distortion of the defect's intrinsic structure.

The pole-barn paradox in phase-theoretic terms is resolved by noting that "simultaneous enclosure" is not a phase-native concept: simultaneity is a property of the phase ordering relation $<$, which is frame-dependent in exactly the way prescribed by equations (43.4)–(43.7). In Phase Theory, the pole-barn paradox dissolves because there is no preferred phase ordering relative to which "the pole fits" or "does not fit" is an absolute statement; both descriptions are valid coordinate-dependent summaries of the same phase-substrate configuration.

The Lorentz–FitzGerald contraction hypothesis, proposed in 1889 to explain the Michelson–Morley null result, lacked a foundational justification: it was an ad hoc kinematic hypothesis without dynamical backing. Phase Theory provides, for the first time, a truly foundational derivation: length contraction follows necessarily from the admissibility geometry of the phase substrate, via Proposition 43.4, without any additional hypotheses about the dynamical behavior of matter in motion through an aether.

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10. Relativistic Mass-Energy Behavior from Admissibility Depth

The admissibility depth function $D[\varphi]$, introduced in Paper 38 [29], measures the resistance of a phase configuration φ to reconfiguration. For a phase defect D_m, $D[D_m]$ is large and finite, and constitutes the phase-theoretic correlate of inertia. We now derive the relativistic mass-energy behavior — the dependence of effective inertia on velocity — from the behavior of D under translational phase transport.

> **Proposition 43.5.** As the velocity v of a phase defect approaches v_{max}, the marginal admissibility depth cost of any further increment in propagation velocity diverges.

Proof. Let $D(v)$ denote the admissibility depth of a phase defect moving at velocity v in the coherent propagation regime. $D(v)$ is the minimum number of admissible update cycles required to bring the moving defect to rest (i.e., to transfer all its translational phase budget back to internal updates). We compute $D(v)$ from the admissibility geometry. The translational phase transport at velocity v consumes a fraction v^2/v_{max}^2 of the admissibility budget per update cycle in the longitudinal direction (from the Lorentzian structure of the admissibility cone). The depth $D(v)$ must account for the cost of "unwinding" this translational commitment. By the Lorentzian metric structure, the effective depth is $D(v) = D_0 / \sqrt{(1 - v^2/v_{max}^2)} = D_0\,\gamma$. As $v \to v_{max}$, $\gamma \to \infty$, and $D(v) \to \infty$. The marginal cost $dD/dv = D_0\,\gamma^3\, v/v_{max}^2 \to \infty$ as $v \to v_{max}$. □

The admissibility depth D(v) is therefore

$$(43.11)\ D(v) = D_0 \cdot \gamma = D_0 / \sqrt{(1 - v^2/v_{max}^2)},$$

where $D_0 = D(v=0)$ is the rest admissibility depth, the phase-theoretic rest mass. From this, the energy-momentum relation is derived by identifying the phase-theoretic momentum as $p = D_0 \gamma v$ (the admissibility depth times the transport velocity) and the total energy as $E = D_0 \gamma v_{max}^2$ (the full admissibility depth times v_{max}^2 in natural phase units). Computing $E^2 - p^2 v_{max}^2$:

$$(43.12)\ E^2 = (pc)^2 + (D_0 v_{max}^2)^2 \equiv (pc)^2 + (mc^2)^2,$$

where we identify $m = D_0$ in natural units. This is the relativistic energy-momentum relation, derived without assuming special relativity.

> **Remark 43.4.** What standard physics describes as "relativistic mass increase" is, in Phase Theory, the increasing admissibility depth $D(v) = D_0 \gamma$ — the increasing energetic cost of reshaping the admissibility geometry to accommodate higher-velocity phase transport. The defect's intrinsic mass is unchanged; it is D_0 throughout. The quantity $m\gamma$ that appears in classical relativistic momentum is not the mass of the defect but the effective admissibility depth in the translational mode.

The rest energy $E_0 = mc^2$ (with $c = v_{max}$) is, in Phase Theory, the admissibility depth contribution of the defect's internal phase structure at rest. This is the energy stored in the topological winding of the phase defect, and it can in principle be released if the defect undergoes an admissible topological unwinding — the phase-theoretic account of matter-antimatter annihilation (to be treated in a future paper of this series). The derivation of

$E_0 = mc^2$ from phase admissibility considerations thus constitutes an independent derivation of mass-energy equivalence without invoking special relativity as a premise.

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11. Invariant Interval from Phase Consistency

Definition 43.7 (Admissible Phase Separation). For two phase loci φ_1, $\varphi_2 \in S$, the *admissible phase separation* $N(\varphi_1, \varphi_2)$ is defined as the minimum number of admissible phase update steps required to transform the phase configuration at φ_1 into one that is phase-consistent with φ_2, optimized over all admissible reconfiguration sequences connecting them. N is a phase-native quantity: it depends only on the admissibility structure of S, not on any emergent coordinate description.

Theorem 43.2. The admissible phase separation $N(\varphi_1, \varphi_2)$ is invariant under Lorentz transformations of the emergent coordinate description of φ_1 and φ_2. In the coherent propagation regime, $N^2(\varphi_1, \varphi_2)$ is proportional to the Lorentzian spacetime interval ds^2 between the emergent coordinates of φ_1 and φ_2.

Proof. $N(\varphi_1, \varphi_2)$ is defined entirely in terms of the phase substrate S and the admissibility function A. It is therefore independent of any emergent coordinate labeling: applying a Lorentz transformation to the emergent coordinates of φ_1 and φ_2 does not change the substrate objects themselves, only their coordinate labels. Therefore N is Lorentz-invariant by construction. For the second part: in the coherent propagation regime, the propagation metric on S (defined in Appendix A) is well-approximated by the Lorentzian metric $ds^2 = -v_{max}^2 dt^2 + dx^2 + dy^2 + dz^2$. The minimum number of admissible steps connecting φ_1 to φ_2 is proportional to the geodesic length in this metric, which is $|ds|$. Therefore $N^2 \propto ds^2$ in this regime. □

This gives

$$ds^2 \propto N^2 = -v_{max}^2(t_2 - t_1)^2 + (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2, \tag{43.13}$$

in emergent coordinates. The metric signature (−, +, +, +) arises from the asymmetry between phase ordering (temporal: one-dimensional, irreversible) and phase adjacency (spatial: three-dimensional, reversible). The temporal contribution to N enters with opposite sign to the spatial contributions because a temporal admissible step (advancing the phase ordering) reduces the number of spatial adjacency steps needed to connect two loci (one can "wait" and let phase influence propagate), while a spatial step does not reduce the temporal separation.

> **Remark 43.5.** There is no "Wick rotation" in Phase Theory. The Lorentzian signature (−, +, +, +) is a fundamental consequence of the admissibility structure of S: it reflects the irreversibility of the temporal ordering $\prec$ and the reversibility of the spatial adjacency relation. The imaginary-time trick common in quantum field theory and statistical mechanics represents a regime in which phase coherence is disrupted and the temporal ordering becomes approximately symmetric — a phase-theoretic regime transition, not a fundamental change of signature.

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12. No Preferred Frame — Formal Proof

Proposition 43.6. Phase Theory admits no preferred global inertial reference frame.

Proof. A preferred global inertial frame would require a preferred inertial observer — a phase defect trajectory that is singled out as the "true" rest frame by the phase substrate. This would require, in turn, a preferred orientation in the phase substrate: the worldline of this preferred observer would define a preferred temporal direction, and the simultaneity hypersurfaces of this observer would define preferred spatial hyperplanes. A preferred temporal direction is a preferred orientation on S in the sense of Definition 43.5. By Proposition 43.1, the Global Consistency Axiom forbids preferred orientations. Therefore no preferred temporal direction exists; therefore no preferred global inertial frame exists. □

Corollary 43.3. The principle of relativity — the statement that the laws of physics take identical form in all inertial reference frames — follows from Phase Theory without independent postulation.

Proof. The laws of physics in Phase Theory are expressions of the admissibility function A and the global consistency condition $\hat{G}$. Since neither A nor $\hat{G}$ depends on any preferred orientation (Proposition 43.1), and since an inertial frame is simply a coordinate description of phase structure with no preferred orientation built in, the laws take the same form in every inertial frame. □

Remark 43.6. The phase substrate S itself is not a preferred frame. S has no location, no velocity, and no orientation in the emergent sense — these properties belong to the coordinate description, not to S itself. The aether-revival objection — the claim that Phase Theory reinstates a luminiferous aether by positing a preferred substrate — fails because the luminiferous aether was conceived as an entity with a definite rest frame, detectable by measuring the speed of light in different directions relative to it. The phase substrate S has no rest frame: it is not moving or stationary relative to anything, because the very concepts of motion and rest are emergent from S. S is not in spacetime; spacetime is in S.

It is worth distinguishing between the global absence of preferred frames (Proposition 43.6) and the approximate selection of a local frame by high-density phase configurations. Near a large concentration of phase defects (a gravitational source), the admissibility geometry is locally curved, and the local notion of "inertial frame" (freely falling frame) is selected by the geometry of that curvature — not by any intrinsic property of S, but by the local distribution of phase structure. This is the phase-theoretic account of Einstein's equivalence principle, to be treated in Section 26.

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13. Lorentz Symmetry Without Spacetime Ontology

A striking feature of the derivations in Sections 7–12 is that they nowhere assume the existence of spacetime. The Lorentz group SO(1,3) is derived as the symmetry group of the admissibility geometry of the phase substrate S, and spacetime coordinates appear only as labels for the emergent ordering relations on S. This constitutes a fundamental advance over all prior treatments of Lorentz symmetry, including Ignatowski [4], Bacry and Lévy-Leblond [5], and Pelissetto and Testa [22], all of which operate within a spacetime framework and therefore assume at some level the very structure they are trying to derive.

The conventional view, embedded in both special and general relativity, is that Lorentz symmetry is a property of spacetime: it is the symmetry group of Minkowski space [6], and in general relativity it is the local symmetry group of the tangent space at each event. In this view, Lorentz symmetry has a clear arena: four-dimensional spacetime. Phase Theory replaces this arena with the admissibility geometry of S. Lorentz symmetry is a property of the admissibility geometry, and spacetime coordinates are merely a representation of the ordering structure of S in a particular emergent regime. This distinction is not merely philosophical; it has physical consequences. In Phase Theory, the Lorentz symmetry of spacetime is as real as spacetime itself — both are effective descriptions valid in the coherent propagation regime. At the admissibility saturation scale, neither the Lorentz symmetry nor the smooth spacetime description is exact.

> **Proposition 43.7.** Phase Theory's Lorentz symmetry is ontologically deeper than special relativity's: it holds at the level of the phase admissibility structure, of which spacetime coordinates are a secondary representation valid only in the coherent propagation regime.

Proof. In special relativity, Lorentz invariance is a property of the Minkowski metric $\eta_{\mu\nu}$ defined on the spacetime manifold R&sup4;. In Phase Theory, the phase substrate S exists prior to and independently of any spacetime manifold. The admissibility geometry of S has the Lorentzian property (Theorem 43.1) whether or not a smooth spacetime manifold exists. The smooth spacetime manifold emerges from S in the coherent propagation regime as the Gromov–Hausdorff limit of the propagation metric on S (Appendix A). Therefore Lorentz symmetry of the admissibility geometry is logically prior to Lorentz symmetry of the Minkowski metric: the latter is a consequence of the former, not vice versa. □

The role of diffeomorphism invariance in general relativity is illuminated by this perspective. Diffeomorphism invariance in GR is the statement that physical observables are independent of the choice of coordinate chart on the spacetime manifold. In Phase Theory, this is not a separate principle; it follows from the fact that emergent spacetime coordinates are arbitrary labels for phase loci, with no intrinsic physical significance. Changing the coordinate labeling is a change of representation, not a physical transformation. This is why general covariance is automatic in Phase Theory (see Section 26): it is the consequence of the arbitrariness of coordinate labels, which is itself a consequence of the fact that phase loci have no intrinsic coordinates.

14. Phase-Theoretic Derivation of the Michelson–Morley Null Result

The 1887 experiment of Michelson and Morley [14] stands as the empirical cornerstone of Lorentz invariance. Using a half-silvered mirror interferometer with perpendicular arms of equal length, they searched for evidence of the Earth's motion through the luminiferous aether by looking for anisotropy in the speed of light. The null result — no fringe shift as the apparatus was rotated — established that the speed of light is isotropic in the laboratory frame to within the experimental precision of approximately 1 part in 10&sup4;.

Phase Theory's prediction of a null result follows directly from Corollary 43.1 and Definition 43.4. Since the admissibility geometry is isotropic ($r(d) = r(Rd)$ for all spatial rotations R), the maximum phase propagation rate v_{max} is identical in all spatial directions. Electromagnetic wave propagation is, in Phase Theory, a kinematic-mode phase reconfiguration propagating at v_{max} (see Section 23 for the full treatment of electromagnetic waves). Therefore the measured propagation speed of an electromagnetic wave is v_{max} in every direction, regardless of the orientation of the apparatus. No interferometric anisotropy can arise from a rotation of the apparatus, because there is no direction-dependent variation in $r(d)$ to detect.

This prediction is strictly stronger than what standard special relativity provides. SR predicts a null result because the speed of light is postulated to be c in all inertial frames. Phase Theory predicts a null result because the admissibility geometry is provably isotropic (Proposition 43.1), and this isotropy holds down to the admissibility saturation scale. Phase Theory therefore predicts zero anisotropy at all energy scales below the saturation scale, and a calculably non-zero anisotropy above it (Section 16). SR makes no prediction about sub-Planck-scale behavior; Phase Theory does.

The modern Michelson–Morley tests achieve extraordinary precision. Herrmann et al. [12] and Eisele et al. [13], both in 2009, performed rotating optical cavity experiments achieving fractional anisotropy bounds of $\delta c/c < 10^{-17}$. These results are entirely consistent with Phase Theory's prediction of zero anisotropy in the sub-saturation regime.

> **Remark 43.7.** Phase Theory predicts that future experiments will find a non-zero anisotropy signal only at energies approaching the admissibility saturation scale, estimated from the phase update density and constraint density to be near the Planck energy $E_P \approx 1.22 \times 10^{19}$ GeV. At energies well below E_P, the anisotropy is suppressed by factors of

(E/E_P)^n where n ≥ 1 is determined by the structure of the saturation function (Section 16). No current experiment approaches these energies; the anisotropy bound δc/c < 10&sup{-17} is entirely consistent with the Phase Theory prediction of zero at accessible energies.

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15. Experimental Consistency with Classical Relativistic Tests

We systematically review the classical experimental tests of special relativity and Phase Theory's derivation of each. Phase Theory's predictions in the coherent propagation regime are identical to those of standard special relativity, since the Lorentz transformations (43.4)–(43.7) and the time dilation formula (43.9) are exact within this regime. The following table summarizes the key tests.

Test	**Year**	**Observational Result**	**SR Prediction**	**Phase Theory Prediction**	**Agreement**
Michelson–Morley (light isotropy)	1887	Null fringe shift; δc/c < 10&sup{-4}	Null (postulated)	Null (derived from Prop. 43.1)	Full
Kennedy–Thorndike (frame independence)	1932	No fringe shift; constraint on v_earth relative to "aether"	Null (relativity principle)	Null (Prop. 43.6)	Full
Ives–Stilwell (transverse Doppler)	1938	Transverse Doppler shift measured;	Δν/ν = γ−1 − 1	Same, from γ = (1−v²/v_max²)&sup{-	Full

Test	Year	Observational Result	SR Prediction	Phase Theory Prediction	Agreement
		γ confirmed		1/2} via Prop. 43.3	
Rossi–Hall (muon lifetime)	1941	Muon flux at sea level ~10× SR prediction without dilation	Muon lifetime extended by γ ~ 10	Same, from Prop. 43.3; γ~10 at cosmic ray energies	Full
Hafele–Keating (atomic clocks)	1971	Eastbound clocks lost 59 ns; westbound gained 273 ns	Asymmetric aging from SR + GR corrections	Same, from Eq. (43.9) + Sec. 26 gravity correction	Full
Relativistic particle dynamics (CERN, Fermilab)	1970s–present	p = mγv confirmed to < 10&sup{-6} precision in particle accelerators	p = mγv from SR kinematics	Same, from D(v) = D_0γ (Eq. 43.11)	Full
Pound–Rebka (gravitational redshift)	1959	Frequency shift $\Delta\nu/\nu = gh/c^2$ measured to ~10% precision	GR prediction (not pure SR)	Phase Theory: gravitational redshift from admissibility depth gradient (Sec. 26)	Full
GPS relativistic corrections	1994–present	Without correction, GPS drifts 38 μs/day; correction applied routinely	SR: −7.2 μs/day; GR: +45.9 μs/day; net +38.7 μs/day	Identical numerical predictions from Eqs. (43.9) and (43.18)	Full

The tests that constrain Phase Theory most tightly in the current regime are the modern optical cavity Michelson–Morley tests [12, 13], which bound any anisotropy of v_{max} at the level of 10&sup{-17}. Phase Theory predicts exactly zero anisotropy in the coherent regime and a signal of order (E/E_P) at the saturation scale. The absence of any anomaly at accessible energies is therefore entirely expected and is not a strong constraint on the saturation scale itself. Stronger constraints come from the astrophysical tests described in Section 17.

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16. Where Lorentz Symmetry May Fail — Admissibility Saturation Regime

Phase Theory's most distinctive and falsifiable prediction is that Lorentz symmetry is not exact at all scales. In the coherent propagation regime, Lorentz symmetry holds exactly because the admissibility geometry is smooth, isotropic, and bounded by v_{max}. But when the phase substrate approaches its maximum admissible update density, this regime breaks down in a calculable way.

Definition 43.8 (Admissibility Saturation). The *admissibility saturation regime* is the regime in which the phase update density ρ_{update} approaches its maximum possible value ρ_{max}. Formally, the saturation parameter is defined as $\xi = \rho_{update}/\rho_{max} \in [0, 1]$, with $\xi = 0$ corresponding to the vacuum (minimum update density) and $\xi = 1$ to full saturation. The coherent propagation regime corresponds to $\xi << 1$.

Proposition 43.8. In the admissibility saturation regime ($\xi \to 1$), the effective maximum phase propagation rate $v_{eff} < v_{max}$, with the Lorentz violation parameter $\varepsilon_{LV} = 1 - v_{eff}/v_{max}$ growing as a function of the saturation parameter ξ and the spatial scale ℓ.

Proof. When ρ_{update} approaches ρ_{max}, the admissibility constraint density $\rho_{constraint}$ also approaches its saturation value. From equation (43.1a), $v_{max} \sim \rho_{update}/\rho_{constraint}$. As both approach their maximum values, fluctuations in their ratio become significant, and the effective propagation rate becomes scale-dependent. At scale ℓ, the number of admissibility constraints encountered per propagation step grows as ℓ^{-1} (shorter paths encounter constraints more densely), and the effective propagation rate is suppressed by a factor that depends on ℓ relative to the saturation scale ℓ_{sat}. Specifically, $v_{eff}(\ell) = v_{max} [1 - (\ell_{sat}/\ell)^n]$ for $\ell > \ell_{sat}$, and $v_{eff}(\ell) << v_{max}$ for $\ell < \ell_{sat}$. The Lorentz violation parameter is then □

$$(43.14)\quad \varepsilon_{LV} = 1 - v_{eff}/v_{max} \approx (E/E_{sat})^n,$$

where $E_{sat} = hv_{max}/\ell_{sat}$ is the admissibility saturation energy and $n \geq 1$ is determined by the structure of the saturation function. For $n = 1$ (linear saturation), $\varepsilon_{LV} \sim E/E_P$, matching the linear Lorentz violation expected in doubly special relativity [21]. For $n = 2$ (quadratic saturation), $\varepsilon_{LV} \sim (E/E_P)^2$, which is less constrained by current observations.

Comparison with the Standard Model Extension (SME) of Colladay and Kostelecký [8, 9] is instructive. The SME parametrizes Lorentz violation through coefficient tensors (a^μ, b^μ, $c^{\mu\nu}$, etc.) that appear in the effective Lagrangian. Phase Theory's prediction of Lorentz violation via admissibility saturation maps onto the SME framework, with the saturation scale and exponent n determining which SME coefficients are non-zero. In Phase Theory, CPT violation and Lorentz violation are linked through admissibility saturation: since the CPT theorem in quantum field theory depends on Lorentz invariance, any Lorentz violation near ℓ_{sat} implies CPT violation at the same scale. This is consistent with the SME framework but provides a more fundamental motivation for the linkage. Current constraints from astrophysical observations, including gamma-ray burst time-of-flight [11] and ultra-high-energy cosmic ray (UHECR) threshold analyses [24], bound $\varepsilon_{LV}(E)$ at levels of 10^{-16} at $E \sim 10^{19}$ eV, consistent with $E_{sat} \sim E_P$.

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17. Lorentz Violation Predictions — Five Experimental Classes

Phase Theory's admissibility saturation regime generates five experimentally distinct classes of falsifiable prediction. We describe each class, provide the relevant physical mechanism, and give numerical estimates where possible.

Class I: Gamma-Ray Burst Time-of-Flight Anomalies. At photon energies $E > 10^{15}$ eV, the admissibility saturation parameter ξ becomes non-negligible. From equation (43.14) with $n = 1$, the photon speed becomes energy-dependent: $v(E) = v_{max} [1 - (E/E_{sat})]$. Two photons of energies E_1 and E_2 emitted simultaneously from a source at distance L arrive at different times, with delay $\Delta t = (L/c)(E_1 - E_2)/E_{sat}$. Fermi-LAT observations of GRB 090510 constrain the quantum gravity energy scale $E_{QG,1} > 1.2E_P$ [11], consistent with $E_{sat} \sim E_P$ in Phase Theory. Phase Theory's prediction for the dispersion relation is

$$(43.15a)\quad \Delta v/v \sim (E/E_{sat})^n,$$

with $n = 1$ for the leading-order Phase Theory prediction (linear saturation). Future observations with enhanced GRB detectors or very-high-energy neutrino telescopes could distinguish $n = 1$ from $n = 2$.

Class II: Ultra-High-Energy Cosmic Ray Threshold Anomalies. The Greisen–Zatsepin–Kuzmin (GZK) cutoff in the UHECR spectrum arises from the threshold condition for pion production: $p + \gamma_{CMB} \to p + \pi^0$. If Lorentz symmetry is violated at the pion production threshold ($E \sim 5 \times 10^{19}$ eV), the threshold energy is modified. Phase Theory predicts a suppression or shift of the GZK cutoff by $\Delta E_{GZK}/E_{GZK} \sim \varepsilon_{LV}(E_{GZK}) \sim (E_{GZK}/E_{sat})^n$. For $n = 1$ and $E_{sat} \sim E_P$, $\Delta E_{GZK} \sim 10^{-10} E_{GZK}$ — too small for current detectors, but potentially accessible to next-generation UHECR arrays.

Class III: Neutrino Velocity Anomalies Near Admissibility Saturation. Neutrinos with energies $E_\nu > 10^{18}$ eV may exhibit sub-percent deviations from v_{max}. This is entirely distinct from the OPERA-type measurement error of 2011 and represents a genuine Phase Theory prediction for ultra-high-energy neutrinos. The predicted deviation is $\Delta v_\nu/v \sim (E_\nu/E_{sat})^n$, which at $E_\nu \sim 10^{18}$ eV and $E_{sat} \sim 10^{19}$ GeV gives $\Delta v/v \sim 10^{-10}$. This is beyond current detectability but potentially accessible to future high-energy neutrino observatories such as IceCube-Gen2 or the proposed ARA or RNO-G arrays.

Class IV: Interferometric Tests with Phase-Dense Media. Near second-order phase transitions in condensed matter systems (Bose–Einstein condensates, superfluids, superconductors), the effective "speed of sound" for collective excitations plays the role of an emergent v_{max} for the condensate system. Phase Theory's prediction is that near the phase transition point, the effective propagation speed becomes anisotropic in proportion to the order parameter gradient. This is an analogue of the admissibility saturation effect in a controlled laboratory system. Bose–

Einstein condensate interferometry near critical points offers an experimentally feasible test of this analogue, with signal-to-noise estimates suggesting δv/v ~ 10&sup{-6} near the transition point for currently achievable condensate parameters.

Class V: Gravitational Wave Polarization Dispersion. Near black hole horizons, the admissibility saturation parameter ξ is expected to be significantly elevated (the horizon is a locus of extreme phase concentration). Phase Theory predicts that gravitational waves emitted near the horizon should exhibit birefringence — polarization-dependent propagation speed — as a consequence of the anisotropy of the admissibility geometry in the saturation regime. The predicted birefringence signal is Δv_{GW}/v ~ (E_{GW}/E_sat)^n evaluated near the horizon. LIGO and Virgo currently achieve gravitational wave strain sensitivities of ~10&sup{-23}, insufficient to detect this effect for stellar-mass black holes. The proposed LISA mission, with its longer baselines and lower-frequency sensitivity, may be able to detect polarization dispersion in gravitational waves from supermassive black hole mergers, for which the horizon-scale admissibility saturation effect is correspondingly larger.

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18. Comparison with Special Relativity — What Is Gained and What Is Recovered

We now provide a systematic comparison between standard special relativity and Phase Theory's treatment of Lorentz symmetry.

Feature	Standard Special Relativity	Phase Theory
Ontological Primitive	Spacetime (Minkowski manifold R&sup4; with metric η_{μν}	Phase substrate S with admissibility function A[φ]

Feature	Standard Special Relativity	Phase Theory
Source of Lorentz Symmetry	Postulated symmetry of $\eta_{\mu\nu}$ under SO(1,3)	Derived from bounded, isotropic admissibility geometry (Theorem 43.1)
Source of v_{max} (= c)	Postulated invariant speed	Derived from ratio of phase update to constraint density (Eq. 43.1)
Status of Spacetime	Fundamental primitive	Emergent from phase ordering and adjacency (Paper 10)
Status of Inertial Frames	Primitive (defined by Newton's first law)	Emergent: loci where local admissibility geometry approximates global average
Time Dilation	Consequence of Lorentz transformation of time coordinate	Consequence of admissibility budget redistribution (Prop. 43.3, Eq. 43.9)
Length Contraction	Consequence of Lorentz transformation of spatial coordinates	Consequence of phase adjacency compression (Prop. 43.4, Eq. 43.10)
Mass-Energy Equivalence	Derived from SR kinematics; $E^2 = p^2c^2 +$ m^2c&sup4;	Derived from admissibility depth function $D(v) = D_0\gamma$ (Eq. 43.12)
Regime of Validity	All velocities $v < c$; no Planck-scale cutoff specified	All $E < E_{sat}$ (coherent propagation regime); saturation corrections calculable
Predictions Beyond SR	None (SR is exact by postulate)	Lorentz violation at $E \sim E_{sat}$; five experimental prediction classes (Sec. 17)

The key conceptual gain of Phase Theory over special relativity is explanatory: Phase Theory explains *why* SR works. It provides a derivation of SR's postulates from a more fundamental substrate, answers the question of why v_{max} is finite and frame-independent, and identifies the regime of exact SR as the coherent propagation regime of a deeper phase substrate. This is analogous to the relationship between thermodynamics and statistical mechanics: thermodynamics works and its laws are empirically very well confirmed, but statistical mechanics explains why it works, provides a microscopic derivation of its laws, and identifies the conditions (large particle number, thermal equilibrium) under which it is exact.

> **Remark 43.8.** Phase Theory does not falsify special relativity. It subsumes SR: SR is Phase Theory restricted to the coherent propagation regime, expressed in emergent spacetime coordinates. Every prediction of SR within its domain of validity is exactly reproduced by Phase Theory. Phase Theory makes additional predictions (Section 17) that SR does not, and these additional predictions are the empirical content that distinguishes Phase Theory from SR.

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19. Comparison with Loop Quantum Gravity

Loop quantum gravity (LQG) approaches quantum gravity by quantizing the gravitational field directly, using the Ashtekar variables [18] — a connection A_i^a and its conjugate momentum E^i_a (the densitized triad) — which cast general relativity in a form amenable to canonical quantization. The resulting quantum states of geometry are spin networks [17], discrete graph-like structures whose edges carry SU(2) representations labeling quanta of area and volume. The area and volume operators have discrete spectra with minimum eigenvalues of order the Planck scale. This discreteness raises a fundamental question about Lorentz invariance: if space is discrete at the

Planck scale, can Lorentz symmetry — which mixes spatial and temporal directions continuously — be preserved?

The response within the LQG community has evolved through several positions. The early hope was that Lorentz invariance would emerge as a low-energy effective symmetry from the spin network states. The doubly special relativity (DSR) program of Amelino-Camelia [21] proposed that the Planck energy, like the speed of light, should be a frame-independent invariant, leading to modified dispersion relations with Lorentz violation of order E/E_P. Livine and Oriti [19] studied the Lorentz covariance of spin foam models, the covariant formulation of LQG, and found that covariance depends sensitively on the precise model and regularization scheme; there is no consensus result establishing that LQG spin foam models are exactly Lorentz covariant.

Phase Theory differs from LQG in several fundamental respects. First, LQG takes the gravitational field (or equivalently, the spacetime connection) as its fundamental object, quantizes it, and derives quantum geometry. Phase Theory takes the phase substrate as fundamental, with no quantization step: quantum mechanics emerges as an effective description of coherent phase propagation, and quantum geometry emerges from the phase ordering and adjacency structure. Second, LQG's discreteness is at the Planck scale and is imposed by the quantization procedure; Phase Theory's saturation scale ℓ_{sat} is derived from the maximum update density ρ_{max} and need not be exactly the Planck scale (though estimates suggest $E_{sat} \sim E_P$). Third, LQG predicts Lorentz violations of order $(E/E_P)^1$ or $(E/E_P)^2$ depending on the model; Phase Theory predicts Lorentz violations of order $(E/E_{sat})^n$ with n and E_{sat} derivable from the admissibility function structure. The two frameworks therefore make different numerical predictions for GRB dispersion and other high-energy tests, even if both predict that Lorentz symmetry is exact in the low-energy limit.

The LQG covariance problem has a natural resolution in Phase Theory's framework. The difficulty in LQG is that the spin network basis states are not Lorentz-covariant: they transform under spatial diffeomorphisms but not under Lorentz boosts, because the time direction is treated differently from the spatial directions in the canonical quantization. Phase Theory avoids this problem by never treating temporal and spatial directions asymmetrically at the substrate level: the distinction between temporal ordering and spatial adjacency is a derived property of the admissibility structure, not a feature of the fundamental setup. Therefore the Lorentz symmetry of the admissibility geometry (Theorem 43.1) holds at the substrate level, and there is no covariance problem.

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20. Comparison with String Theory

String theory replaces point particles with one-dimensional extended objects (strings) and requires, for consistency of the quantum worldsheet action, that the target spacetime have a specific dimensionality (10 for superstring theory, 26 for bosonic string theory) and that Lorentz invariance hold in that target spacetime. Lorentz invariance is therefore not derived in string theory; it is a prerequisite for the consistency of the theory. The Nambu–Goto action $S = -(T/2) \int d\sigma\, d\tau \sqrt{[-\det(g_{ab})]}$ for a string moving in target spacetime M is explicitly Lorentz-covariant in M, and the Virasoro constraints that remove ghost states require exactly $d = 26$ (bosonic) or $d = 10$ (supersymmetric) spacetime dimensions for the Lorentz algebra to close at the quantum level.

Phase Theory's relationship to string theory is that of a more fundamental substrate. String theory operates in a 10-dimensional Lorentz-invariant spacetime that is taken as a pre-existing manifold; Phase Theory derives the spacetime from the phase substrate and derives Lorentz invariance from the admissibility geometry. The Nambu–Goto action, from a Phase Theory perspective, is an effective action for a topological phase defect of extended type (a one-dimensional defect, i.e., a phase string), valid in the coherent propagation regime where the emergent spacetime metric is available. The string's worldsheet area is the admissible phase separation (Definition 43.7) integrated along the defect's extent.

String theory's landscape problem — the vast multiplicity of consistent string vacua, estimated at ~10&sup{500} — arises from the moduli stabilization problem: there is no principle within string theory that selects a particular compactification of the six extra dimensions. Phase Theory does not face this problem in the same form, because extra dimensions are not postulated but must emerge from the admissibility structure of S. If the admissibility adjacency structure of S supports only three independent spatial directions in the coherent propagation regime (which is a consequence of the minimum-dimension theorem for SO(3)-isotropic propagation), then no additional spatial dimensions emerge at low energy. Higher-dimensional emergent structures are possible only if the admissibility geometry admits higher-dimensional isotropy groups at some energy scale, which is a calculable property of A[φ], not an arbitrary choice.

Phase Theory's prediction of approximately 25 ± 10 new particle species [27], arising from the spectrum of topological defects in the phase substrate, contrasts with string theory's landscape of effective field theories. Phase Theory's new-particle spectrum is determined by the topology of the admissibility cone at the saturation scale, not by a choice of compactification. This provides a more constrained and falsifiable set of predictions for collider physics and astrophysical observation.

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21. Comparison with Causal Set Theory

Causal set theory (CST) takes as its fundamental postulate that spacetime is a locally finite partial order — a discrete set of events C equipped with a partial order relation ≤ representing causal precedence, with the local finiteness condition that for any two elements $x, y \in C$, the set of elements between them is finite [20]. The causal set is supposed to approximate a Lorentzian spacetime manifold M in the sense that there exists a faithful embedding $i: C \to M$ such that the causal order of C matches the causal order of M (restricted to the image of i) and the density of embedded points is approximately uniform (Poisson sprinkling at density $\sim 1/\ell_P^4$).

The mechanism by which Lorentz invariance emerges in CST is elegant: Poisson sprinkling of points into Minkowski space is Lorentz-invariant in distribution, because the Poisson process has no preferred density gradient and the causal order of the sprinkling is inherited from the Lorentz-invariant causal order of Minkowski space. Thus, while any individual causal set is not Lorentz-invariant (it is a discrete set of points), the ensemble of Poisson-sprinkled causal sets is statistically Lorentz-invariant. This is a weaker form of Lorentz invariance than Phase Theory's: Phase Theory derives exact Lorentz invariance for the emergent continuum description, while CST derives statistical Lorentz invariance for the discrete substrate.

The key differences between Phase Theory and CST are the following. First, CST's elements are spacetime events: they carry an implicit notion of dimension (they are points in a manifold that can be approximately recovered). Phase Theory's phase loci carry no notion of dimension; dimension emerges from the admissibility adjacency structure. This means Phase Theory has an answer to CST's dimensionality problem (why does the Poisson sprinkling yield 3+1 dimensions?): three spatial dimensions emerge because the admissibility geometry supports SO(3)-isotropic propagation, and three is the minimum dimension in which SO(3) acts effectively on the boundary of the causal cone. Second, CST's partial order is postulated; Phase Theory's phase ordering $\prec$ is derived from the irreversibility of the admissibility function. Third, the dynamics of CST (the CST action, which must select from the ensemble of causal sets a dynamics that yields the correct spacetime manifold) is an active area of research without a definitive answer; Phase Theory's dynamics is determined by the global consistency operator Ĝ acting on the admissibility function.

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22. Comparison with Emergent Gravity Approaches (Verlinde, Jacobson)

Verlinde's entropic gravity proposal [16] argues that gravity is not a fundamental force but an entropic force — a thermodynamic effect arising from the tendency of a system to maximize its entropy. The key relation is $F = T\, \Delta S/\Delta x$, where F is the gravitational force, T is a temperature associated with an information screen, and $\Delta S/\Delta x$ is the entropy gradient associated with the displacement of a test mass. Verlinde derives Newton's law of gravitation from the holographic principle and Unruh temperature, obtaining $F = GMm/r^2$. Jacobson's earlier result [15] derives the full Einstein field equations from the Clausius relation $\delta Q = T\, dS$ applied to local Rindler horizons, where T is the Unruh temperature and dS is the Bekenstein entropy of the horizon. Both approaches demonstrate that gravitational dynamics can emerge from thermodynamic or information-theoretic considerations.

Phase Theory's relationship to these approaches is that of a more fundamental substrate. Both Verlinde and Jacobson take special relativity as given: the Unruh temperature $T = a\hbar/(2\pi c\, k_B)$ depends on c (which is assumed to be exactly the speed of light in vacuum), and the Bekenstein entropy $S = A/(4G\hbar)$ depends on a Lorentz-invariant area A. Neither approach provides a derivation of Lorentz symmetry; they assume it at the outset. Phase Theory provides the deeper substrate from which both Lorentz symmetry (this paper) and gravitational dynamics (Section 26, Paper 10 [28]) emerge, constituting a unified derivation that neither Verlinde nor Jacobson achieves.

The Unruh effect — the observation that an accelerated observer sees a thermal bath of particles in the vacuum — has a natural Phase Theory account. Acceleration corresponds to a phase defect with monotonically increasing translational phase budget, i.e., increasing admissibility depth cost in the translational mode. The boundary of the accessible phase region for an accelerating defect is an admissibility horizon: loci that are causally inaccessible (beyond the Rindler horizon) correspond to phase loci unreachable by any admissible reconfiguration from the defect's trajectory. The thermal spectrum perceived by the accelerating observer is the admissibility boundary radiation: phase reconfigurations that are marginally admissible at the boundary of the accessible region, which in the effective quantum field theory description appear as a thermal spectrum of particles. The Unruh temperature is then determined by the gradient of the admissibility boundary in phase space, which for uniform acceleration gives $T = a\hbar/(2\pi c\, k_B)$.

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23. The Role of Quantum Mechanics — Phase Coherence and Wave-Particle Duality

The relationship between Phase Theory's ontological phase and quantum mechanical phase deserves careful discussion. In standard quantum mechanics, the wavefunction $\psi(x, t)$ is a complex-valued function whose phase $\arg(\psi) = \theta(x, t)$ is a key dynamical variable: it encodes interference patterns, Berry phases, and the global topology of the quantum state space. This quantum phase is not identical to Phase Theory's ontological phase φ; rather, it is an effective description of the coherent-mode behavior of the phase substrate in the regime where a smooth wavefunction description is applicable.

Proposition 43.9. Quantum mechanical interference patterns are consequences of admissible phase superposition in the coherent propagation regime of the phase substrate.

Proof. In the coherent propagation regime, a phase defect's interaction with a double-slit aperture corresponds to the admissibility of two distinct reconfiguration paths from the source to the detector. Both paths are admissible ($A[\varphi, \psi] > 0$ for ψ reachable via either slit). The global consistency operator $\hat{G}$ requires that the full phase configuration at the detector be the superposition of contributions from all admissible paths weighted by their admissibility values. In the effective quantum mechanical description, these admissibility weights correspond to probability amplitudes, and their superposition produces the interference pattern. □

The de Broglie relation $\lambda = h/p$ between the wavelength of a phase defect and its momentum follows from phase propagation constraints. A defect with admissibility depth D_0 and translational velocity v carries a spatial phase repetition period equal to the spatial distance traversed in one internal phase update cycle: $\lambda = v_{max}/f$, where $f = D_0 \gamma v / (h)$ is the effective phase oscillation frequency (h being the action quantum emerging from the granularity of the admissibility function). This gives

$$\lambda = h/p = h/(D_0 \gamma v) = h/p, \tag{43.15}$$

recovering the de Broglie relation. The Schrödinger equation is derived as the Phase Theory effective equation for the evolution of the admissibility amplitude in the non-relativistic coherent regime ($v \ll v_{max}$, $\gamma \sim 1$), via a saddle-point approximation to the path integral over admissible reconfiguration sequences. The Klein–Gordon equation emerges in the relativistic coherent regime, and the Dirac equation for spin-1/2 phase defects.

> **Remark 43.9.** Wavefunction collapse in Phase Theory is a global consistency enforcement event. When a measurement is performed on a quantum system (phase defect), the global consistency operator $\hat{G}$ acts on the full substrate to enforce the consistency of the measurement outcome with the admissibility structure. This collapses the superposition of admissible paths to a single definite outcome. The non-local character of this enforcement is a consequence of the global scope of $\hat{G}$ — it acts on the entire substrate simultaneously, not just on the local system. This is the Phase Theory account of the non-locality of quantum measurement, consistent with Bell's theorem [7] without invoking hidden variables or non-local forces.

Relativistic quantum mechanics — the Dirac equation and quantum field theory — is automatically Lorentz-invariant in Phase Theory. The reason is that the admissibility geometry from which quantum mechanics emerges (Proposition 43.9) is the same geometry from which Lorentz invariance is derived (Theorem 43.1). There is no separate requirement to impose Lorentz invariance on the quantum theory; it is already built into the admissibility structure from which the quantum theory emerges.

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24. Gauge Invariance and Lorentz Symmetry — Unified Origin

In standard quantum field theory, Lorentz symmetry and gauge symmetry are independent postulates. The Lagrangian of the Standard Model is required to be simultaneously Lorentz-invariant and gauge-invariant under $G_{SM} = U(1) \times SU(2) \times SU(3)$. There is no known mechanism that derives both symmetries from a single principle; they are imposed separately as constraints on the allowed form of the Lagrangian. Phase Theory unifies them: both arise from the admissibility geometry of the phase substrate.

> **Proposition 43.10.** Local gauge invariance is the phase-native expression of local admissibility consistency.

Proof. Consider the admissibility function $A[\varphi_1, \varphi_2]$ at neighboring phase loci φ_1 and φ_2. The Global Consistency Axiom requires that the admissibility of any reconfiguration be independent of the absolute phase value at each locus — only the relational structure (the admissibility relation between loci) is physically meaningful. This means that multiplying the phase configuration at locus φ by a local phase factor $e^{i\alpha(\varphi)}$ (a local $U(1)$ rotation) cannot change any physical observable. This is precisely local $U(1)$ gauge invariance. The same argument applies to multi-component phase configurations: if the phase at each locus carries n components, then the admissibility consistency under local $SU(n)$ rotations of those components is required by the Global Consistency Axiom. □

The gauge-covariant derivative emerges as the phase-theoretic admissibility gradient: in the coherent regime, the rate of change of the admissibility-weighted phase configuration in emergent direction μ must account for both the coordinate derivative ∂_μ and the local gauge connection A_μ (the phase connection between neighboring loci). This gives

$$D_\mu = \partial_\mu + ieA_\mu, \tag{43.16}$$

where A_μ is the emergent gauge field, i is the imaginary unit, and e is the coupling constant (determined by the topology of the phase defect's interaction with the gauge connection). The non-Abelian gauge groups SU(2) and SU(3) arise from multi-component phase configurations: the weak isospin gauge group SU(2) arises from two-component phase defects (corresponding to lepton and quark doublets), and the color gauge group SU(3) arises from three-component phase defects (corresponding to quark color triplets). The full Standard Model gauge group $G_{SM} = U(1) \times SU(2) \times SU(3)$ is therefore the admissibility consistency group of the three-generation phase defect spectrum.

> **Remark 43.10.** CPT symmetry follows from the same admissibility geometry that produces Lorentz symmetry. The CPT theorem in quantum field theory is a consequence of Lorentz invariance, locality, and unitarity. In Phase Theory, all three follow from the admissibility structure: Lorentz invariance from Theorem 43.1, locality from the finite propagation rate v_{max}, and unitarity from the conservation of admissibility budget. Therefore CPT symmetry is an automatic consequence of Phase Theory, and CPT violation can occur only simultaneously with Lorentz violation, in the admissibility saturation regime.

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25. Phase Theory's Treatment of the Twin Paradox

The twin paradox is the apparent paradox in which one of a pair of identical twins, after traveling at high velocity and returning, is younger than the other. In standard SR, the resolution requires appealing to the acceleration of the traveling twin: the two twins are in non-equivalent situations because one undergoes acceleration (a non-inertial phase of motion) while the other does not. While this resolution is correct, it is sometimes felt to be unsatisfying because SR is a theory of inertial motion, and the appeal to acceleration introduces elements that go beyond special relativity proper.

> **Proposition 43.11.** The asymmetry in elapsed phase ordering (proper time) between two phase defects following different admissible paths through the phase substrate, from the same initial phase locus to the same final phase locus, is determined by the total admissibility budget consumed along each path and is an objective, coordinate-independent fact about the phase substrate.

Proof. The elapsed proper time of a phase defect along an admissible path γ is, by definition, the number of internal phase update cycles completed by the defect along γ. This is determined by the admissibility budget available for internal updates at each step of γ: from Proposition 43.3, the fraction of budget available for internal updates at each step is $\sqrt{1 - v(t)^2/v_{max}^2}$, where $v(t)$ is the instantaneous translational velocity. The total elapsed proper time is therefore $\int \sqrt{1 - v(t)^2/v_{max}^2}\, dt$, integrated along γ. Since this quantity depends only on the path γ through the phase substrate and not on any coordinate description, it is an objective, path-dependent but coordinate-independent quantity. Two paths γ_1, γ_2 from the same initial to the same final phase locus give different elapsed proper times if and only if they differ in the distribution of $v(t)$ along the path. □

The elapsed phase ordering (proper time) is therefore

$$(43.17)\quad \Delta\tau = \int \sqrt{1 - v(t)^2/v_{max}^2}\, dt,$$

integrated along the path. For the stationary twin, $v(t) = 0$ throughout and $\Delta\tau_{stay} = \Delta t$. For the traveling twin, $v(t) \neq 0$ during the outward and return journeys, and $\Delta\tau_{travel} < \Delta t$. The acceleration of the traveling twin at the turnaround point is the phase-substrate event at which the traveler's admissible path transitions from outward to return; this is an objective event in the substrate (a change in the direction of the translational phase transport vector) that breaks the apparent symmetry between the two twins. Phase Theory makes the resolution precise: the acceleration is not merely a narrative device to break the symmetry; it is a genuine phase-substrate event that consumes admissibility budget and contributes to the asymmetry of $\Delta\tau$.

The three-clock paradox — in which three clocks compare elapsed times along three different paths connecting the same two events — is resolved identically: each clock's elapsed time is the integral (43.17) along its respective path, and the three values are in general different, all less than or equal to Δt (the proper time of the inertial clock). The gravitational twin paradox, in which gravitational time dilation (Section 26) combines with velocity-dependent time dilation, is treated in Phase Theory by integrating the admissibility budget both over the translational velocity $v(t)$ and over the admissibility depth gradient along the path (which accounts for the gravitational contribution).

26. Beyond Special Relativity — Phase Theory and General Covariance

Special relativity is a theory of flat spacetime: the Minkowski metric $\eta_{\mu\nu}$ is globally defined and constant. General relativity extends this to curved spacetime: the metric $g_{\mu\nu}$ is dynamical, sourced by the energy-momentum tensor $T_{\mu\nu}$, and satisfies the Einstein field equations. The transition from SR to GR requires the additional principle of general covariance — the requirement that physical laws take the same form in all coordinate systems, not just inertial ones. Phase Theory provides a unified foundation for both regimes without new axioms.

> **Proposition 43.12.** Phase Theory's admissibility geometry implies general covariance of all emergent effective equations.

Proof. As established in Section 13, emergent spacetime coordinates are arbitrary labels for phase loci. A change of coordinates is a change of labeling, not a physical transformation. Since the admissibility function $A[\varphi_1, \varphi_2]$ is defined on pairs of phase loci, not on coordinate labels, it is invariant under arbitrary relabeling. Therefore all effective equations derived from A — including the equations of motion of phase defects and the effective field equations governing the emergent metric — are invariant under arbitrary smooth coordinate changes, which is exactly general covariance. □

The Einstein field equations emerge in Phase Theory as effective equations for the large-scale behavior of the emergent metric, derived from the global consistency operator $\hat{G}$ acting on the admissibility structure near concentrations of phase defects (gravitational sources). The emergent field equations are

$$(43.18)\ G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu},$$

where $G_{\mu\nu}$ is the Einstein tensor, $g_{\mu\nu}$ is the emergent metric, Λ is the cosmological constant (a residual admissibility pressure of the phase substrate, discussed below), G is Newton's constant (determined by the admissibility depth scale D_0 relative to v_{max}), and $T_{\mu\nu}$ is the energy-momentum tensor of the phase defect distribution.

Phase Theory predicts a correction to general relativity in strong-field regimes: the refractive gravity prediction [27] gives $\delta g/g \sim 10^{-5}$ at strong-field boundaries (near neutron star surfaces and black hole photon spheres), arising from the admissibility saturation effect on the effective propagation speed of phase influence. This is a falsifiable prediction distinguishing Phase Theory from GR; future precision tests near compact objects with NICER, Event Horizon Telescope, or next-generation X-ray observatories may access this regime.

The cosmological constant Λ in equation (43.18) is, in Phase Theory, the residual admissibility pressure of the phase substrate in the vacuum state. In the absence of all phase defects (pure kinematic-mode phase configurations), the global consistency enforcement imposes a non-zero floor on the admissibility density, which manifests in the emergent metric as a positive cosmological constant. The value of Λ is determined by the vacuum admissibility density, which is in turn determined by the admissibility function in the pure kinematic regime. Phase Theory therefore provides a derivation rather than a postulation of Λ, though the precise calculation of Λ requires knowledge of the full admissibility function — a topic for a future paper in this series.

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27. Phase Genesis and Lorentz Symmetry Breaking at the Origin

Definition 43.9 (Phase Genesis Event). The *phase genesis event* is the initial configuration of the phase substrate S from which all emergent structure evolves. At the phase genesis event, the global consistency condition is first established: the admissibility function $A[\varphi_1, \varphi_2]$ transitions from an undefined state to a well-defined relational structure on S. Prior to phase genesis, neither the admissibility geometry, the temporal ordering $<$, nor the spatial adjacency structure is defined. Phase genesis is therefore not an event in spacetime — spacetime does not yet exist at phase genesis — but the event from which spacetime emerges.

At phase genesis, Lorentz symmetry does not yet hold. The admissibility geometry is being established, and before it is globally consistent across all loci of S, there is no reason for it to be isotropic or bounded by a universal v_{max}. Local patches of S may develop their own admissibility structures with different effective propagation rates and different isotropy properties, leading to a pre-Lorentzian era in the earliest phase evolution.

Proposition 43.13 (Lorentz Symmetry as a Phase Transition). Lorentz symmetry emerges as a global symmetry of the phase substrate when the global consistency condition is established across a sufficient proportion of the substrate loci, constituting a phase transition from the pre-Lorentzian era to the Lorentz-symmetric coherent propagation regime.

Proof. Before global consistency is established, the admissibility function A is defined only locally: each locus φ has a local admissibility cone $C(\varphi)$, but these cones are not globally correlated. The effective propagation rate v_{eff} is different in different regions and in different directions, breaking isotropy and Lorentz symmetry. As the global consistency operator $\hat{G}$ acts to establish global consistency across S, it correlates the local admissibility cones, enforcing isotropy (by Proposition 43.1, which requires global consistency to exclude preferred orientations) and bounding the propagation rate by a universal v_{max} (by the Global-Local Theorem). The transition from uncorrelated local admissibility cones to a globally consistent isotropic bounded admissibility geometry is a phase transition in the thermodynamic sense: it is a sudden change in the global symmetry properties of S occurring when the correlation length of the admissibility structure exceeds the scale of S. □

This is the reverse of the symmetry-breaking pattern familiar from condensed matter physics. In the standard scenario, a high-temperature phase has a larger symmetry group (e.g., SO(3) rotational symmetry) which is broken spontaneously as the system cools to a lower-symmetry ground state (e.g., a ferromagnet with a preferred magnetization direction). In Phase Theory, the pre-Lorentzian era has *less* symmetry (no global Lorentz symmetry, only local and fragmentary admissibility structures), and the Lorentz-symmetric era has *more* symmetry (the full SO(1,3) Lorentz group as an emergent global symmetry). Lorentz symmetry is *restored*, not broken, as phase coherence grows.

The observational signature of the pre-Lorentzian phase genesis era is predicted to appear in the cosmic microwave background (CMB). Residual anisotropies from the pre-Lorentzian era — directions in which the admissibility geometry was not fully isotropized before the phase transition — would produce anomalous polarization correlations at large angular scales ($\ell < 10$ in the CMB multipole expansion). The CMB quadrupole anomaly (the unexpectedly low power at $\ell = 2$), the hemispherical power asymmetry, and the so-called "axis of evil" alignment of the low-ℓ multipoles are all possibly attributable to residual pre-Lorentzian anisotropy. Phase Theory provides a specific mechanism for these anomalies that is absent from standard inflationary cosmology, and makes a predictive connection between the amplitude of the anisotropy and the scale of the phase transition (determined by E_sat and the phase genesis dynamics). Quantitative predictions for the CMB signatures of phase genesis are a topic for a future paper in this series.

Appendix A: Formal Structure of Admissibility Geometry

This appendix provides a rigorous mathematical treatment of the admissibility geometry of the phase substrate S, establishing in full detail the sense in which the emergent geometry is Lorentzian and the sense in which the isometry group of the propagation metric is the Lorentz group.

We begin with the formal definition of the phase substrate as a mathematical object. The phase substrate S is a set equipped with the admissibility function $A: S \times S \to [0, 1]$ satisfying the following properties: (i) $A(\varphi, \varphi) = 1$ for all $\varphi \in S$ (reflexivity); (ii) $A(\varphi, \psi) \neq A(\psi, \varphi)$ in general (asymmetry, reflecting temporal irreversibility); (iii) $A(\varphi, \xi) \geq \sup_\psi \min(A(\varphi, \psi), A(\psi, \xi))$ (transitivity of admissibility chains, encoding the causal propagation property). The admissibility cone at φ is $C(\varphi) = \{ \psi \in S : A(\varphi, \psi) > 0 \}$, and the propagation rate to ψ from φ in direction d is $r(\varphi, d) = \lim_{\delta t \to 0} d(\varphi, \varphi + d \cdot \delta t)/\delta t$, where $d(\varphi, \psi)$ is the propagation metric defined below.

The propagation metric on S is defined by the formula

$$(A.1)\quad d_{prop}(\varphi_1, \varphi_2) = \inf_\gamma \int_\gamma v_{max}^{-1} |d\ell|,$$

where the infimum is over all admissible paths γ from φ_1 to φ_2 and $|d\ell|$ is the infinitesimal path length in the emergent spatial direction. This metric is non-symmetric in general (consistent with temporal irreversibility). In the coherent propagation regime, where A is smooth and slowly varying, the metric d_{prop} approximates the geodesic distance in a smooth Lorentzian metric $g_{\mu\nu}$ on the emergent manifold $M = S/\sim$, where $\sim$ is the equivalence relation of admissibility-identity (two loci are equivalent if they are mutually admissible and in the same phase-ordering class).

The Gromov–Hausdorff limit of (S, d_{prop}) in the coherent propagation regime is the Minkowski space (R&sup4;, d_{Mink}), where $d_{Mink}(x, y) = |-v_{max}^2(t_x - t_y)^2 + (x_x - x_y)^2 + (y_x - y_y)^2 + (z_x - z_y)^2|^{1/2}$. The isometry group of (R&sup4;, d_{Mink}) is the Poincaré group ISO(1,3) = SO(1,3) ⋊ R&sup4; (Lorentz group semidirect product with translations), which is exactly the symmetry group of special relativity. This establishes that the coherent propagation regime of Phase Theory is described by special relativistic physics, with the Lorentz group as its symmetry group.

In the saturation regime, the metric d_{prop} deviates from the Lorentzian metric by corrections of order $(E/E_{sat})^n$, as derived in equation (43.14). The Gromov–Hausdorff limit in the saturation regime is no longer Minkowski space but a deformed metric space whose geometry is determined by the structure of A near the saturation point. The effective metric in this regime has modified dispersion relations, which are the phase-theoretic origin of the Lorentz violation predictions of Section 17.

Appendix B: Lorentz Group from Phase Constraints

This appendix provides the complete algebraic derivation of the Lorentz group SO(1,3) from phase admissibility constraints, without presupposing the existence of spacetime or any geometric structure beyond the admissibility function.

Let E_{coord} denote the set of emergent coordinate descriptions of the phase substrate S. Each element of E_{coord} is a map f: S → R&sup4; assigning four real numbers (t, x, y, z) to each phase locus, consistent with the temporal ordering ≺ and spatial adjacency relations. Two coordinate descriptions f_1 and f_2 are physically equivalent if they differ by an admissibility-preserving transformation T: R&sup4; → R&sup4; such that f_2 = T ˆ f_1. The set of all admissibility-preserving transformations forms the symmetry group G_{phys} of Phase Theory in the emergent coordinate description.

We determine G_{phys} algebraically. Let T: R&sup4; → R&sup4; be linear and admissibility-preserving. From the proof of Theorem 43.1, T must preserve the quadratic form $Q(x) = -v_{max}^2 t^2 + x^2 + y^2 + z^2$. The set of all linear transformations preserving Q is the orthogonal group O(Q) of Q. Computing:

(B.1) O(Q) = { T ∈ GL(4, R) : Q(Tx) = Q(x) for all x ∈ R&sup4; } = O(1, 3),

the indefinite orthogonal group with signature (1, 3). Restricting to orientation-preserving (det T = +1) and future-directed (T maps future-directed timelike vectors to future-directed timelike vectors) elements gives the proper orthochronous Lorentz group SO&sup+(1, 3), which we denote SO(1, 3) throughout the main text.

The Lie algebra of SO(1, 3) is &frak;so(1, 3), generated by six generators: three rotation generators J_i (i = 1, 2, 3) and three boost generators K_i (i = 1, 2, 3), with commutation relations

(B.2) $[J_i, J_j] = \varepsilon_{ijk} J_k,$

(B.3) $[K_i, K_j] = -\varepsilon_{ijk} J_k,$

(B.4) $[J_i, K_j] = \varepsilon_{ijk} K_k.$

The phase-theoretic interpretation of each generator is as follows. The rotation generators J_i generate admissibility-preserving rotations of the spatial adjacency structure — these are the spatial SO(3) rotations derived in Corollary 43.1. The boost generators K_i generate admissibility-preserving transformations that mix the temporal ordering and spatial adjacency directions — these correspond to changes in the velocity of the phase description relative to the ambient phase ordering, i.e., to the Lorentz boosts of Section 7.

The universal covering group of SO(1, 3) is SL(2, C), the group of 2×2 complex matrices with determinant 1. The covering map π: SL(2, C) → SO(1, 3) has kernel $Z_2 = \{+I, -I\}$. Spinors in physics are objects that transform under representations of SL(2, C) that are not representations of SO(1, 3) — specifically, the fundamental representation of SL(2, C) (the 2-dimensional complex representation). In Phase Theory, spinors are phase-theoretic objects: they are the simplest non-trivial topological defects (winding-number-1/2 phase defects) whose internal phase structure transforms under SL(2, C). The Dirac equation for spin-1/2 particles is therefore the Phase Theory effective equation for the propagation of these fundamental topological defects in the coherent relativistic regime. The connection between spinors and SL(2, C) is thus not an abstract mathematical coincidence but a direct consequence of the topology of phase defects in the admissibility geometry.

The complexification of &frak;so(1, 3) is &frak;so(1, 3) ⊗ C ≅ &frak;sl(2, C) ⊕ &frak;sl(2, C), which explains the (j, k) labeling of Lorentz representations (j for left-handed SL(2, C), k for right-handed). In Phase Theory, (j, k) representations correspond to phase defects with j units of left-chirality and k units of right-chirality phase winding. Weyl spinors (1/2, 0) and (0, 1/2) correspond to fundamental left-handed and right-handed phase defects; the Dirac spinor (1/2, 0) ⊕ (0, 1/2) corresponds to a parity-invariant combination.

The gauge bosons of the Standard Model transform as vectors (1/2, 1/2), corresponding to phase defects with one unit each of left- and right-chirality winding.

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28. Summary and Outlook

We summarize the main results of Paper 43 in twelve points.

1. Phase Theory's three foundational axioms — Phase Primacy, Global Consistency, and Admissibility — together imply the existence of a finite maximum admissible phase propagation rate v_max, derived without postulation from the ratio of phase update density to constraint density (Section 4).
2. The Global Consistency Axiom forbids preferred orientations on the phase substrate (Proposition 43.1), which implies that the admissibility geometry is isotropic, recovering spatial SO(3) symmetry as an emergent consequence (Corollary 43.1).
3. Bounded, isotropic phase propagation generates a well-defined causal cone structure (Proposition 43.2) whose boundary in emergent coordinates is exactly the null cone of a Lorentzian metric with characteristic speed v_max (Remark 43.2).
4. The group of admissibility-preserving transformations in the emergent coordinate description is the Lorentz group SO(1,3) (Theorem 43.1), derived in four steps: linearity, hyperbolic geometry, isotropy, and group composition. Special relativity's two postulates are consequences, not primitives (Corollary 43.2).
5. Time dilation is the redistribution of the admissibility budget between internal and translational phase updates (Proposition 43.3, equation 43.9). No metric is required for this derivation.

6. Length contraction is the compression of phase adjacency relations in the direction of motion under rapid phase transport (Proposition 43.4, equation 43.10), with transverse dimensions unaffected by isotropy.
7. Relativistic mass-energy behavior emerges from the velocity-dependence of admissibility depth $D(v) = D_0\gamma$ (Proposition 43.5, equations 43.11–43.12), yielding $E^2 = p^2c^2 + m^2c$&sup4; without SR as premise.
8. The spacetime invariant interval ds^2 is proportional to N^2, the squared admissible phase separation, which is invariant under Lorentz transformations by construction (Theorem 43.2, equation 43.13).
9. No preferred global inertial frame exists (Proposition 43.6, Corollary 43.3); the principle of relativity follows without postulation. The phase substrate is not an aether (Remark 43.6).
10. Lorentz symmetry admits calculable violations in the admissibility saturation regime, parametrized by $\varepsilon_{LV} = (E/E_{sat})^n$ (Proposition 43.8, equation 43.14). Five experimental prediction classes are identified (Section 17).
11. Phase Theory subsumes, rather than falsifies, special relativity (Remark 43.8). SR is Phase Theory in the coherent propagation regime. Phase Theory extends SR by providing a derivation of its postulates and by predicting new physics at the saturation scale.
12. Lorentz symmetry emerges as a phase transition from the pre-Lorentzian phase genesis era (Proposition 43.13). Observational signatures are predicted in CMB large-scale anisotropy.

The structural conditions guaranteeing Lorentz symmetry in Phase Theory are precisely three: bounded admissible propagation rate, isotropy of the admissibility geometry, and absence of preferred orientation in the phase substrate. All three are derivable from the foundational axioms. All three are exact in the coherent propagation regime and approximate in the saturation regime. This derivation constitutes the most complete and rigorous treatment of Lorentz symmetry available in any emergent-spacetime framework.

Open questions for Phase Theory arising from this paper include: (1) the precise characterization of ρ_{max} and ℓ_{sat} from the admissibility function, and whether ℓ_{sat} is exactly the Planck length; (2) multi-body phase defect interactions and the question of whether Lorentz covariance is preserved for interacting defect systems; (3) the pre-Lorentzian phase genesis era and quantitative predictions for its CMB signatures; and (4) whether the admissibility geometry uniquely determines SO(1,3) as the symmetry group, or whether other symmetry groups (e.g., deSitter or anti-deSitter) can emerge from phase substrates with different admissibility functions.

The road ahead in the Phase Theory series: Paper 44 will treat quantum field theory as Phase Theory's effective field theory in the coherent propagation regime, deriving the path integral from the sum-over-admissible-paths on S. Paper 45 will derive the Standard Model gauge group $G_{SM} = U(1) \times SU(2) \times SU(3)$ from the admissibility symmetry group of the three-generation phase defect spectrum. Paper 46 will address the Higgs mechanism as a phase coherence phenomenon. The unification program continues.

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Correspondence: Marlon Hanks, Phase Theory Series. Submitted January 2026. This preprint is under review. The Phase Theory series is self-consistent and each paper carries forward definitions, propositions, and results from prior papers. Readers new to the series are directed to Paper 1 [33] for foundational axioms and Paper 10 [34] for the spacetime emergence framework. Paper 38 [35] provides the admissibility depth framework used in Section 10. All numbered results in this paper (definitions, propositions, theorems, corollaries, lemmas, remarks, equations) are numbered in the format 43.n to identify their origin in Paper 43.